

15. Which of the following statement is *not* true about closed sets ?

- (A) Arbitrary union of closed sets is closed.
- (B) Arbitrary intersection of closed sets is closed.
- (C) If $X = (x_n)$ is a sequence of elements in a closed set F , then $\lim X$ belongs to F .
- (D) A subset of $\mathbb{R}$ is closed if and only if it contains all of its cluster points.

16. Which of the following statements is *not* true about Cantor set ?

- (A) Cantor set is closed.
- (B) Cantor set is uncountable..
- (C) The complement of Cantor set in $[0, 1]$ has length 1.
- (D) Cantor set has non-empty open intervals as subsets.

17. If $z = (x, y)$ is a complex number its inverse z^{-1} is :

- (A) $\left(\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2} \right)$.
- (B) $\left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right)$.
- (C) $\left(\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$.
- (D) $\left(\frac{y}{x^2 + y^2}, \frac{-x}{x^2 + y^2} \right)$.

18. $(1 - i)^4$ is equal to .

- (A) 4.
- (B) $4i$.
- (C) -4 .
- (D) $-4i$.

19. If $z = \frac{-2}{1 + \sqrt{3}i}$, then $\text{Arg } z$ is :

- (A) $\frac{\pi}{3}$.
- (B) $-\frac{\pi}{3}$.
- (C) $\frac{2\pi}{3}$.
- (D) $-\frac{2\pi}{3}$.

20. $|e^{i\theta}|$ is equal to :

- (A) $\sqrt{2}$.
- (B) $-\sqrt{2}$.
- (C) 1.
- (D) -1 .

**FIFTH SEMESTER U.G. (CUCBCSS—UG) DEGREE [SPECIAL]
EXAMINATION, NOVEMBER 2020**

Mathematics

MAT 5B 07—BASIC MATHEMATICAL ANALYSIS

Time : Three Hours

Maximum : 120 Marks

Section A

*Answer all questions.
Each question carries 1 mark.*

1. Fill in the blanks : Infimum of the $S = \{1 - 1/m; m \in \mathbb{N}\}$ is _____.
2. The Set of all real numbers which satisfy the inequality $|x^2 - 1| \leq 3$ is _____.
3. Fill in the blanks : The ε neighbourhood of $b \in \mathbb{R}$ is _____.
4. State the Supremum Property of $\mathbb{R}$.
5. State Cantor's theorem.
6. Fill in the blanks : If X is a converging sequence of non-negative real numbers, then $\lim X =$ _____.
7. Define Cauchy sequence.
8. State the Betweenness property of irrational numbers.
9. State the Monotone Sub-sequence Theorem.
10. Give an example of a bounded real sequence which is not a Cauchy sequence.
11. Fill in the blanks : The value of $(1+i)^{10} + (1-i)^{10}$ is _____.
12. Fill in the blanks : $\text{Arg}(-\pi - i\pi) =$ _____.

(12 × 1 = 12 marks)

Section B

*Answer at least eight questions.
Each question carries 6 marks.
All questions can be attended.
Overall Ceiling 48.*

13. Define Supremum and Infimum of set. Find them for the Set $S = \left\{1 - \frac{(-1)^n}{n}; n \in \mathbb{N}\right\}$.

Turn over

14. Prove that the set $\mathbb{N}$ of positive integers is not bounded above.
15. If b is a real number such that $0 \leq b \leq \varepsilon$ for every $\varepsilon > 0$, then prove that $b = 0$.
16. Show that there doesn't exist a rational number r such that $r^2 = 7$.
17. State and prove Squeeze theorem on sequence.
18. Test the convergence of the sequence $\left(\frac{\log n}{n}\right), \dots$
19. If X and Y are convergent sequences of real numbers satisfying $(x_n) \leq (y_n), \forall n \in \mathbb{N}$, then prove that $\lim(x_n) \leq \lim(y_n), \forall n \in \mathbb{N}$.
20. Prove that every bounded sequence of real numbers has a converging sub-sequence.
21. Illustrate Cauchy sequence by an example.
22. Distinguish between converging sequence and Cauchy sequence. Write the relation between them.
23. Prove or disprove that "the union of infinitely many closed sets in $\mathbb{R}$ is closed".
24. Prove that $\|z_1\| - \|z_2\| \leq \|z_1 - z_2\|$.
25. Test the convergence of the sequence $\left(\frac{\cos n}{n}\right)$.
26. Find the principal value of $(-125i)^{\frac{1}{3}}$.

(8 × 6 = 48 marks)

Section C

*Answer at least five questions.
Each question carries 9 marks.
All questions can be attended.
Overall Ceiling 45.*

27. State and prove Characterization theorem of Intervals.
28. Prove that $[0, 1]$ is uncountable.
29. State and prove the "Density theorem".
30. Determine the set $A = \left\{x \in \mathbb{R} : \frac{2x+1}{x+2} < 1\right\}$.
31. Let $X = (x_n)$ be a non-negative sequence of real numbers with $\lim(x_n) = x$. Prove that $\lim(\sqrt{x_n}) = x$.

32. (a) Give an example of a convergent sequence (x_n) of positive real numbers with $\lim(x_n)^{\frac{1}{n}} = 1$.
- (b) Give an example of a divergent sequence (x_n) of positive real numbers with $\lim(x_n)^{\frac{1}{n}} = 1$.
- (c) Justify the property of the sequence (x_n) of positive real numbers with $\lim(x_n)^{\frac{1}{n}} = 1$.
33. Test whether the (x_n) defined by $x_n = 1 + 1/2 + 1/3 + \dots + 1/n$ is Cauchy sequence or not.
34. Establish the relation between contractive sequence and convergent sequence.
35. Give an algebraic proof for the triangle inequality of complex numbers.

$$|(z_1 + z_2)| \leq |(z_1)| + |(z_2)|, \forall z_1, z_2 \in \mathbb{C}.$$

(5 × 9 = 45 marks)

Section D

*Answer any one question.
The question carries 15 marks.*

36. (a) Distinguish between countable and uncountable sets. Give examples for each of them.
- (b) Prove that :
- (i) Subset of a countable set is countable.
- (ii) Superset of an uncountable set is uncountable.
37. State and prove the Cauchy convergence criteria for a sequence.
38. (a) Define closed sets in $\mathbb{R}$. Show that the Intersection of an arbitrary collection of closed sets in $\mathbb{R}$ is closed.
- (b) Show by an example that the union of infinitely many closed sets in $\mathbb{R}$ need not be closed.

(1 × 15 = 15 marks)

**FIFTH SEMESTER U.G. (CUCBCSS—UG) DEGREE [SPECIAL]
EXAMINATION, NOVEMBER 2020**

Mathematics

MAT 5B 06—ABSTRACT ALGEBRA

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 06—ABSTRACT ALGEBRA

(Multiple Choice Questions for SDE Candidates)

1. Which of the following defines a binary operation on $\mathbb{Z}^+$?
 - (A) $a * b = a - b$.
 - (B) $a * b = c$, where c is the smallest integer greater than both a and b .
 - (C) $a * b = c$, where c is at least 5 more than $a + b$.
 - (D) $a * b = c$, where c is the largest integer less than the product of a and b .
2. If b and c are the inverses of some element a in a group G then :
 - (A) $b = c$.
 - (B) $b \neq c$.
 - (C) $b = kc$ for some $k \in \mathbb{N}$.
 - (D) None of these.
3. On $\mathbb{Q}$, which of the following does not define a binary operation ?
 - (A) $a * b = |a| |b|$.
 - (B) $a * b = (a - b)^2$.
 - (C) $a * b = +\sqrt{ab}$.
 - (D) None of these.
4. Let $*$ be the binary operation defined on $\mathbb{Q}^+$ as $a * b = ab/2$. Then inverse of the element a is :
 - (A) $2a$.
 - (B) $4/a$.
 - (C) a^2 .
 - (D) None of these.
5. Which of the following are true ?
 - (1) A group may have more than one identity element.
 - (2) Any two groups of three elements are isomorphic.
 - (3) Every group of at most three elements is abelian.
 - (A) 2 and 3.
 - (B) 1 and 2.
 - (C) 1 and 3.
 - (D) All.
6. Let $G = \{1, -1, i, -i\}$ where, be a set of four elements. Which of the following is a binary operation on G ? (1) $a * b = a + b$. (2) $a * b = a.b$.
 - (A) Only 1.
 - (B) Only 2.
 - (C) Both.
 - (D) None of these.

7. In a group G , $(a * b)^2 = a^2 * b^2$ for all $a, b \in G$. This statement is :
- (A) Always true.
 (B) True if G is finite.
 (C) True if G is a multiplicative group.
 (D) True if G is abelian.
8. If a group G is of order 31, then which of the following is false ?
- (A) G is abelian. (B) G is cyclic.
 (C) G is abelian but not cyclic. (D) Both abelian and cyclic.
9. The Klein 4-group is isomorphic to _____.
- (A) $Z_2 \times Z_4$. (B) $Z_2 \times Z_2$.
 (C) Z_4 . (D) None of these.
10. Order of $(2, 2)$ in $Z_4 \times Z_6$ is _____.
- (A) 2 (B) 6.
 (C) 4. (D) 12.
11. Which of the following is true ?
- (A) Every cyclic group has a unique generator.
 (B) In a cyclic group, every element is a generator.
 (C) Every cyclic group has at least two generators.
 (D) None of these.
12. Which of the following is a cyclic group with only one generator ?
- (A) Z_2 . (B) $(Z, +)$.
 (C) Klein-4 group. (D) None of these.
13. Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}$. Then σ^6 equals :
- (A) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}$ (B) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix}$.
 (C) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}$ (D) None of these.

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021**(CUCBCSS—UG)****Mathematics****MAT 5B 06—ABSTRACT ALGEBRA****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 30 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 06—ABSTRACT ALGEBRA

(Multiple Choice Questions for SDE Candidates)

1. Which of the following defines a binary operation on $\mathbb{Z}^+$?
 - (A) $a * b = a - b$.
 - (B) $a * b = c$, where c is the smallest integer greater than both a and b .
 - (C) $a * b = c$, where c is at least 5 more than $a + b$.
 - (D) $a * b = c$, where c is the largest integer less than the product of a and b .
2. Under which of the following binary operation, the set of positive integers is closed ?
 - (A) $b \ a * b = a/b$.
 - (B) $a * b = a - b$.
 - (C) $a * b = ab^2$.
 - (D) $a * b = +\sqrt{ab}$.
3. The order of the subgroup of $\mathbb{Z}_4$ generated by the element 3 is :
 - (A) 3.
 - (B) 4.
 - (C) 2.
 - (D) 1.
4. The set $\{1,2,3,4\}$ is an abelian group under the operation _____.
 - (A) Addition.
 - (B) Addition modulo 5.
 - (C) Multiplication modulo 4.
 - (D) Multiplication modulo 5.
5. Which of the following are true ?
 - 1 A group may have more than one identity element.
 - 2 Any two groups of three elements are isomorphic.
 - 3 Every group of at most three elements is abelian.
 - (A) 2 and 3.
 - (B) 1 and 2.
 - (C) 1 and 3.
 - (D) All.
6. Let $G = \mathbb{C}^*$ (non-zero complex numbers). Which of the following are subgroups of G ?
 - 1 All purely imaginary complex numbers under multiplication.
 - 2 All complex numbers with absolute value 1 under multiplication.
 - (A) Only 1.
 - (B) Both.
 - (C) Only 2.
 - (D) None of these.

7. Which of the following is not true ?
- (A) $(\mathbb{Z}, +)$ is a proper subgroup of $(\mathbb{R}, +)$.
 - (B) $(\mathbb{Q}^+, \cdot)$ is a proper subgroup of $(\mathbb{R}^*, \cdot)$.
 - (C) $(3\mathbb{Z}, +)$ is a proper subgroup of $(\mathbb{Z}, +)$.
 - (D) $(\mathbb{Q}^+, \cdot)$ is a proper subgroup of $(\mathbb{R}, +)$.
8. The Klein 4- group is isomorphic to _____.
- (A) $\mathbb{Z}_2 \times \mathbb{Z}_4$.
 - (B) $\mathbb{Z}_2 \times \mathbb{Z}_2$.
 - (C) $\mathbb{Z}_4$.
 - (D) None of these.
9. Order of $(2, 2)$ in $\mathbb{Z}_4 \times \mathbb{Z}_6$ is _____.
- (A) 2.
 - (B) 6.
 - (C) 4.
 - (D) 12.
10. The center of an abelian group G is :
- (A) $\{e\}$.
 - (B) G .
 - (C) A cyclic subgroup.
 - (D) None of these.
11. If a, b are elements of a group G of order m then order of ab and ba are :
- (A) Same.
 - (B) Equal to m .
 - (C) Unequal.
 - (D) None of these.
12. Number of elements in the cyclic subgroup of the group of non-zero complex numbers under multiplication, generated by, $(1+i)/\sqrt{2}$ is :
- (A) 4.
 - (B) 8.
 - (C) 6.
 - (D) None of these.
13. Which of the following is not true ?
- (A) Every cyclic group of order > 2 has at least two distinct generators.
 - (B) If every proper subgroup of G is cyclic, then G itself is cyclic.
 - (C) Every sub-group of a cyclic group is cyclic.
 - (D) None of these.

14. The number orbits of the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 7 & 1 & 4 & 6 & 3 & 5 \end{pmatrix}$ is _____.

- (A) 1. (B) 2.
(C) 3. (D) 4.

15. The product $(1\ 3\ 6)(2\ 4)$ of two permutation is :

- (A) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 6 & 2 & 5 & 1 \end{pmatrix}$. (B) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 2 & 5 & 4 & 6 \end{pmatrix}$.
(C) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 5 & 6 & 1 & 4 \end{pmatrix}$. (D) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 5 & 4 & 6 & 1 \end{pmatrix}$.

16. Order of the permutation $(4\ 5)(2\ 3\ 7)$ in S_7 is :

- (A) 2. (B) 3.
(C) 5. (D) 6.

17. Which of the following is not true ?

- (A) Every cycle is a permutation. (B) A_3 is a commutative group.
(C) A_5 has 120 elements. (D) None of these.

18. Index of the subgroup $5Z$ in $(Z, +)$ is :

- (A) 3. (B) 5.
(C) 7. (D) Infinite.

19. Suppose G is a group of order 29. Then which of the following is true ?

- (A) G is not abelian.
(B) G has no subgroup other than (e) and G .
(C) There is a group H of order 29 which is not isomorphic to G .
(D) G is a subgroup of a group of order 30.

20. The index of the subgroup $\langle 3 \rangle$ in Z_{24} :

- (A) 3. (B) 6.
(C) 2. (D) 4.

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021

(CUCBCSS-UG)

Mathematics

MAT 5B 06—ABSTRACT ALGEBRA

Time : Three Hours

Maximum : 120 Marks

Part A

*Answer all questions.**Each question carries 1 mark.*

1. The smallest non abelian group has ——— number of elements.
2. The order of the identity element in any group G is ———.
3. State True or False. "Every abelian group is cyclic".
4. State True or False. "Every group of order 31 is cyclic".
5. Give an example of non-cyclic group with four elements.
6. The total number of subgroups of Z_{12} is ———.
7. What are the orbits of the identity permutation σ of a set A ?
8. How many zero divisors are there for the field Z_7 ?
9. How many units are there for the field Z_7 ?
10. Give an example of integral domain which not a field.
11. State True or False. Z is a sub field of Q .
12. Write the number of generators of the group Z_5 under addition modulo 5.

(12 × 1 = 12 marks)

Part B

*Answer any ten questions.**Each question carries 4 marks.*

13. Show that left and right cancellation holds in a group G .
14. Let G be a group and suppose that $a * b * c = e \forall a, b, c \in G$. Show that $b * c * a = e$.
15. Prove that a group G has exactly one idempotent element.
16. Can the identity element be a generator of a cyclic group ?
17. Prove that every cyclic group is abelian.

Turn over

18. Consider the group $\mathbb{Z}_{12}$, under the operation addition modulo 12. Find the order of the cyclic subgroup generated by $3 \in \mathbb{Z}_{12}$.
19. Show that the permutation $(1, 4, 5, 6)(2, 3, 1, 5)$ is an even permutation.
20. What is the order of the cycle $(1, 4, 5, 7)$ in S_8 ?
21. Find the partition of the group $\mathbb{Z}_6$, under the operation addition modulo 6, into cosets of the subgroup $H = \{0, 3\}$.
22. Consider the rings $\langle \mathbb{Z}, +, \cdot \rangle$ and $\langle 2\mathbb{Z}, +, \cdot \rangle$. Verify whether the map $\phi: \mathbb{Z} \rightarrow 2\mathbb{Z}$ defined by $\phi(x) = 2x \forall x \in \mathbb{Z}$ is a ring homomorphism or not.
23. Find the number of generators of the cyclic group of order 8.
24. Solve the equation $x^2 - 5x + 6 = 0$ in $\mathbb{Z}_{12}$.
25. Consider the following two binary structures :
 - (a) $\mathbb{Z}$, the set of integers with ordinary addition ; and
 - (b) $2\mathbb{Z} = \{2n | n \in \mathbb{Z}\}$ the set of even integers with ordinary addition.

Show that the above two binary structures are isomorphic.

26. Let n be a positive integer. Give an example of a group containing n elements.

(10 × 4 = 40 marks)

Part C

Answer any six questions.

Each question carries 7 marks.

27. Let $*$ be defined by $\mathbb{Q}^+$ by $a * b = \frac{ab}{2}$. Show that $(\mathbb{Q}^+, *)$ is an abelian group.
28. Let G be a group. For all $a, b \in G$, prove that $(a * b)^{-1} = b^{-1} * a^{-1}$.
29. Prove that a necessary and sufficient condition that a non-empty subset H of a group G is a subgroup of G is that $a \in H, b \in H \Rightarrow ab^{-1} \in H$.
30. Show that the subgroups of $\mathbb{Z}$ under addition are precisely the groups $n\mathbb{Z}$ under addition for $n \in \mathbb{Z}$.
31. Show that any permutation of a finite set of at least two elements is a product of transpositions.
32. Show that a homomorphism ϕ of a group G is a one-to-one function if and only if $\text{Ker } \phi = \{e\}$.
33. Show that cancellation law holds in a ring R if and only if R has no zero divisors.

34. M_2 denotes the ring of all 2×2 matrices of the form $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where a, b, c, d are rational numbers. Is M_2 a field? Justify your answer.
35. Prove that any integral domain D can be enlarged to (or embedded in) a field F such that every element of F can be expressed as a quotient of two elements of D .

(6 × 7 = 42 marks)

Part D

*Answer any two questions.
Each question carries 13 marks.*

36. Show that subgroup of a cyclic group is cyclic.
37. (a) Define the term orbit, cycle and transposition with respect to a permutation.
- (b) Write the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 4 & 3 & 1 \end{pmatrix}$ as product of disjoint cycles.
- (c) Define even and odd permutation. Write $(1, 4, 5, 6)(2, 1, 5)$ as a product of transpositions.
38. Show that in the ring $\mathbb{Z}_n$, the divisors of 0 are precisely those elements that are not relatively prime to n also show that $\mathbb{Z}_p$ is a field.

(2 × 13 = 26 marks)

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021**(CUCBCSS—UG)****Mathematics****MAT 5B 05 VECTOR CALCULUS****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 30 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.